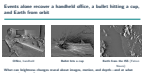
Slide 1. Purpose / pace: Opening; about 30 seconds.

Talking points: The motivating question is how to estimate images, motion, and depth when the sensor reports changes rather than conventional image frames. The mathematical thread is the interaction of observation null spaces, regularization, and identifiability.

Explanation / transition: The talk first introduces the sensor without assuming a computer-vision background, then develops the three inverse problems. The title slide gives the semester.

Events alone recover a handheld office, a bullet hitting a cup, and Earth from orbit



Slide 2. Purpose / pace: Motivation; about 90 seconds. Show the results before any mathematics.

Talking points: Office: a handheld camera in an office; the desk objects and the person are recognizable (about 2.5 million events). Cup: a bullet strikes a ceramic cup from the left; the event stream resolves the fracture and spray, which a standard-rate frame camera would blur (about 750 thousand events). ISS: the Falcon Neuro event cameras on the International Space Station viewing the eastern coastline of Honduras; land–water boundaries and clouds are visible although the scene changes slowly and the events are sparse (about 2 million events). All three come from events alone, with one per-pixel solve and no intensity frames.

Explanation / transition: Advance to the same scenes in false color, then give the motivation there.

Events alone recover a handheld office, a bullet hitting a cup, and Earth from orbit

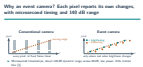


Slide 3. Purpose / pace: Same scenes in false color; about 45 seconds.

Talking points: All three panels now show the same reconstructed frames in false color; grayscale and false color are display choices for the reconstructed relative brightness.

Explanation / transition: Why it matters: fast scenes, wide brightness ranges, and platforms with tight power budgets such as drones and spacecraft. On those platforms the estimator must also be cheap and predictable, which the reconstruction chapter quantifies. Transition: what the sensor reports, and why that makes these problems mathematically interesting.

Why an event camera? Each pixel reports its own changes, with microsecond timing and 140 dB range

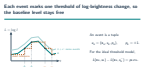


Slide 4. Purpose / pace: Accessibility primer; about two minutes. Assume no event-camera background.

Talking points: Point to the left panel: a conventional camera samples every pixel at the same frame times, whether or not anything changes, and averages over its exposure. Point to the right panel: the same moving bright bar produces a brightening event when its leading edge reaches a pixel and a darkening event when its trailing edge leaves. This is why event displays often look like moving outlines: a boundary passing across a pixel changes its brightness, while a uniform region undergoing the same translation produces little activity. The sensor responds to temporal brightness change; illumination changes can also trigger events. The survey figures quoted on the slide are the standard sensor-class numbers from Gallego et al. (TPAMI 2022).

Explanation / transition: Give the committee the one idea the whole talk uses: events record when the log-brightness at a pixel changed by a fixed amount, so the data determine changes and leave the starting brightness free. Then turn to the next slide for the exact threshold rule. If asked about hardware: the dissertation's MVSEC, office, and cup data come from DAVIS346 sensors, and the orbital data from the Falcon Neuro instrument on the ISS.

Each event marks one threshold of log-brightness change, so the baseline level stays free



Slide 5. Purpose / pace: Sensor primer; 90 seconds.

Talking points: Follow the solid curve. A positive event is emitted after an upward change of one threshold from the most recent reset; a negative event follows a downward change. Different pixels have independent reset times, so the observation times are asynchronous. Neuromorphic here refers to the change-triggered sensing architecture; the estimators are variational models.

Explanation / transition: The equality is an ideal noiseless threshold model. Real threshold uncertainty, missed events, refractory effects, and overshoot create discrepancies. The diagram is conceptual, with arbitrary axes. A stationary bright pixel produces no events: zero activity means no change, at any brightness.



Slide 6. Purpose / pace: Accessibility primer; 90 seconds. Introduce the three estimated outputs.

Talking points: Left: temporal image reconstruction; grayscale is the reconstructed value. Middle: optical-flow estimate, with color encoding direction and displacement magnitude at pixels that recorded events. Right: stereo depth; the two event cameras on a drone give the pair. All three show the same indoor flying 2 scene. Present the panels as the three estimation targets. Explain the brightness baseline, motion along an edge, and stereo correspondence.

Explanation / transition: If asked which methods: Gaussian kernel ($\sigma = 0.1\,\mathrm{s}$) with sigmoid weights; contrast maximization, whose nonconvex objective lies outside the quadratic family analyzed later (existence result in backup); weighted least squares. Reconstruction: 47.690 s. Flow: 47.592–47.719 s; color magnitude represents displacement over this window. White flow pixels have zero events: the mask shows event support. WLS: center 47.598 s, 250 ms window. Different windows and stereo rectification permit scene-level comparison, with separate pixel geometries. Depth colors clip to 0.5–8 m; white masks nonfinite values and the maximum finite depth. Reconstruction grayscale uses a histogram/percentile scale. Exact timestamps, normalization, and provenance are in the adjacent metadata file.

Three estimators, each well posed



Slide 7. Purpose / pace: What the talk shows; about 90 seconds.

Talking points: Each column answers the unobserved quantity named on the previous slide. Reconstruction gets most of the time: the model, its uniqueness and stability, and its cost. Optical flow follows with the main theorem. Stereo is a two-slide summary.

Explanation / transition: The common question links the columns: the events determine some quantities, and the model supplies the rest. The results that go with each column are slides 12 (cost), 20 (stripes and checkerboard), and 22 (MVSEC stereo). The footer tracks the position along this route. Start with the sensor's observation model, which is where the reconstruction begins.

Events fix each pixel's brightness increments and leave one constant free



Slide 8. Purpose / pace: Temporal observation model; 75 seconds.

Talking points: Suppress the pixel index to keep the notation readable. The algorithm chooses the reconstruction times t_i; the penalty on the next slides chooses the reference level. Integrating a unit-mass kernel across a bin assigns all or part of each event to that bin. Delta binning has no spread; Gaussian, rectangular, and spline kernels distribute influence over time. The entries of A are unscaled increments: A f lists the differences of consecutive samples.

Explanation / transition: If Af=0, every consecutive pair of components is equal, hence f is constant. Conversely A1=0. Under ideal threshold calibration the observed increments approximate changes in L/c_th, with kernel-boundary, overshoot, and between-event endpoint errors. No absolute frame is used to set the baseline. Scene terms: each pixel's brightness history is rebuilt on its own; a pixel without events stays at the reference level.

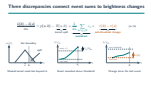
Each kernel spreads an event's sign over nearby bins and conserves its total



Slide 9. Purpose / pace: Kernel choice; 75 seconds.

Talking points: The four kernels are the ones evaluated in Chapter 2: delta (binning), truncated and renormalized Gaussian, B-spline, and rectangular. Each has unit mass, is a translate $\phi_s(t) = \phi_0(t - s)$, and has compact support; the curves share one support width w for display. Unit mass means each event contributes exactly its sign in total; translation invariance means the bin integrals are a convolution; compact support means each event touches finitely many bins. The sum identity is exact: summing the bin integrals telescopes to the integral over the whole window, which is one when the event's full support lies in the window.

Explanation / transition: The plotted widths are matched for display only; the evaluated parameters differ by kernel (Ch. 2 bump sweep). If asked which kernel is best: the indoor and outdoor rankings differ (discussion page on kernel performance), and the uniqueness theorem holds for every kernel because it depends only on **A** and positive weights. Transition: next, when these integrated observations approximate a physical brightness change.



Slide 10. Purpose / pace: Sensor-to-observation model; about 90 seconds. Answers "why is $\mathbf{Af} \approx \mathbf{y}$ a model of an event camera?" before it is asked.

Talking points: Identity first: event sum plus three corrections, one per panel. Then the panels, left to right. Kernel spill: a positive event inside the bin spreads mass beyond b. Its contribution to B is negative; subtracting B restores the signed count. Overshoot: the recorded event exceeds the previous reset plus one threshold. The marked excess is $\varepsilon_\ell / c_{\mathrm{th}}$ on the normalized brightness axis. Endpoint residual: the gap since the last reset is $r(b)$; the identity uses $r(b) - r(a)$. Kernel spill comes from the representation and bin boundaries. The other terms connect event measurements to physical brightness.

Explanation / transition: Assumptions (P1)–(P4): unit-mass kernels, kernels localized relative to event spacing, $|\varepsilon_\ell| \ll c_{\mathrm{th}}$, and positive intensity. A missed or spurious event makes some ε_ℓ comparable to c_{th}, which is outside (P3). The regularized solution adds a fourth discrepancy, the data residual: $f_k^* - f_i^* = y(t_i, t_k) + \sum_{\ell=i}^{k-1} \rho_{\mathrm{res},\ell}$, and larger weights enlarge it. Subthreshold signals can be invisible over a whole window: discussion page "The physical event stream loses more than one baseline". The measure-valued continuous-time limit is the discussion page on the jump primitive. Transition: the events fix changes, so a penalty must choose the level, which is the next theorem.



Slide 11. Purpose / pace: Theorem and stability; about 3 minutes.

Talking points: The objective balances fidelity to the observed changes against a weighted magnitude penalty. Its quadratic form is $\|\mathbf{A}\mathbf{z}\|^2 + \|\mathbf{\Lambda}\mathbf{z}\|^2 \geq \lambda_{\min}^2 \|\mathbf{z}\|^2$, so the normal matrix is positive definite and the minimizer is unique. Subtracting the normal equations for two data vectors gives $\delta\mathbf{f}^* = \mathbf{K}_{\text{time}}^{-1}\mathbf{A}^\top\delta\mathbf{y}$ and, with $\|\mathbf{A}\|_2 \leq 2$, the stated bound.

Explanation / transition: One spurious or missed event changes the data by at most one in norm, because each kernel splits a single unit of polarity across the bins. Premultiplying the normal equations by the constant vector gives the weighted zero-mean identity: the penalty selects the baseline and also changes the fitted increments. The derivation and the condition-number bound are in the backup "Why uniqueness and stability hold".

Each pixel costs one tridiagonal sweep; streaming adds constant work per new frame

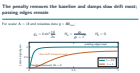


Slide 12. Purpose / pace: Cost structure; about 90 seconds. This slide answers the pre-defense question about cost; Backup 84 gives the per-core timings.

Talking points: The first-difference matrix makes $\mathbf{A}^\top\mathbf{A}$ the path-graph Laplacian. Adding a positive diagonal preserves the tridiagonal structure and gives strict diagonal dominance. One forward and one backward sweep cost O(N t) per pixel, hence O(H W N t) overall. In the implemented sigmoid/delta path, the forward recurrence gives the newest endpoint of a growing prefix. The terminal diagonal is $1 + \lambda^2_{N_t-1}$, because the last interval weight is repeated; the next interval turns that endpoint into an interior row. Three recurrence scalars plus the current output are stored per pixel.

Explanation / transition: The main Python solver uses dense matrices; the edge path exploits the tridiagonal structure. The MVSEC line checks the edge solver against the MATLAB reference. Next: what the penalty does to each temporal mode.

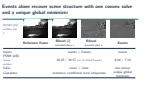
The penalty removes the baseline and damps slow drift most; passing edges remain



Slide 13. Purpose / pace: Spectral interpretation; 75 seconds.

Talking points: The first-difference Gramian is the Neumann-chain Laplacian. Its constant eigenvector has eigenvalue zero. Expanding a noiseless latent signal in the eigenbasis turns reconstruction into multiplication by g/(g+lambda^2). These are latent-signal gains; the gain from noise in the difference observations is a different expression in the backup.

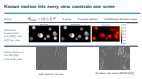
Explanation / transition: The spectral decomposition is proved in Chapter 4 and reused here for stereo. The plotted smooth curves interpolate the finite-chain modal gains. Only omega_k=pi k/n are exact finite-grid modes. This magnitude penalty attenuates low temporal frequencies most, whereas the temporal kernels may average rapid changes. The penalty selects the baseline that the events leave free. Scene terms: slow brightness drift fades and passing edges remain, hence the edge-enhanced look.



Slide 14. Purpose / pace: MVSEC evidence and the bilevel relation; about 2.5 minutes.

Talking points: Top: one MVSEC outdoor frame, then the two bilevel outputs Antil–Sayre define, then the event-only reconstruction. The bilevel reconstruction $v = d - g(z)$ starts from the log frame and corrects its blur, and this frame is one it was fitted to, so v closely matches it. The dynamic part u is the event-driven component; it shares the event-only panel's display scale, so these two compare directly. Both recover the tree, the parked vans, and the hood glare; u also carries texture from the frames, and the event-only brightness is the level the model selects. This chapter: events only, then activity weights, then one fixed convex solve. Antil–Sayre: events and frames, with an outer optimization over $z_{xy} \in \mathbb{R}^{n_t}$, which parameterizes the event-based blur kernel $g(z)$ that links each blurry frame to the latent image. PSNR is similarity to a reference frame in dB; the bilevel PSNR is measured on the frames it is fitted to, so it shows how closely v reproduces its own input frames.

Explanation / transition: PSNR compares with a reference frame, so it also scores the absolute brightness the model selects; the events constrain only changes. The methods receive different inputs, including intensity frames for the comparator, so the table is descriptive: the frames supply the bilevel PSNR. On frames it was not fitted to, the bilevel scores 24.3 / 25.2 dB, matching interpolation between the camera frames (24.3 / 25.2 dB). Kernel comparisons are in the backup "Temporal evidence: events fix structure; the penalty sets the baseline".



Slide 15. Purpose / pace: Shared-scene bridge; about 90 seconds. Read top to bottom: model, test with known truth, real data.

Talking points: Orbit hook first: the ISS moves at about 7.7 km/s, so frame exposures smear the ground; events give microsecond timing, about 140 dB, changes only, for a low-power payload. Known orbit: every view sees the same ground. Model lane: P_k maps the scene θ into view k, A differences views. Simulation lane: the error runs along rows, the weakly constrained direction for horizontal motion. ISS lane: per-pixel structure is faint; pooling views brings out the coastline. Chapter 3 makes the motion the unknown.

Explanation / transition: If asked for the algebra (Proposition 2.4.1):
$\ker B_{\mathrm{SAR}} = \{\theta : P_{k+1}\theta = P_k\theta\}$ contains constants because $P_k\mathbf{1} = \mathbf{1}$; under translation, fields constant along the motion are weakly constrained. With unsquared ridge $\lambda > 0$, $(B^\top B + \lambda I)\widehat{\theta} = B^\top b$ has a unique solution and $\|\delta\theta\| \le \|B_{\mathrm{SAR}}\|\,\|\delta b\|/\lambda$. Orbital panel: L^2 penalty, native 240×180 grid, one-pixel-shift bins; disks: smoothed TV, offset removed. JIF in Chapter 4 applies the same shared-unknown idea to two cameras.

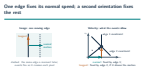
Three different motions look identical through a small window



Slide 16. Purpose / pace: Why events for motion, then the aperture problem visually; about 2 minutes.

Talking points: Motivation: a drone needs image motion now. Frames at 30–60 Hz compare two pictures; a fast edge jumps and blurs, and the answer arrives a frame late. Events timestamp the edge as it crosses each pixel; the time-surface slope gives its normal speed ($\nabla \mathcal{T} \cdot \mathbf{v} = 1$). Optical flow is apparent velocity in the image plane. An aperture simply means a restricted observation window. First look at the circular windows: the moving stripes look alike. Revealing the full patterns shows three different translation directions. A local straight edge tells us its speed perpendicular to the edge; motion along its tangent stays free.

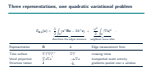
Explanation / transition: Play the masked panel first, then the revealed panel. Under steady illumination, different velocities with the same normal component produce the same local brightness changes and ideal events. Smoothness couples neighboring velocity estimates, while additional observed directions constrain the component left free by a single edge. Next, draw the same ambiguity for one edge and state the flow model in scene terms.



Slide 17. Purpose / pace: Flow model in scene terms; about two minutes.

Talking points: A pixel fires when its log brightness changes by the threshold, so firing times trace an image edge. Left: normal motion in teal, tangent in amber. Right, velocity space, no positions: each line is every velocity one edge's events allow. Teal is the normal part edge 1 measures; amber is the tangent part it leaves free. If edge 2 shares the motion, the lines cross at $\mathbf{v}$, teal plus amber. Shared motion is an assumption: non-parallel lines always cross, so for edges moving differently the crossing is wrong for both. Requirement: two orientations somewhere.

Explanation / transition: With exact coefficients and one shared velocity, the constant field is the unique minimizer at any α and separation. Example: edge 1 vertical moving right at s, edge 2 with normal $(1,1)/\sqrt{2}$ stationary; estimate $(s,-s)$ everywhere. Time-surface coefficient at an edge: rank one, proportional to $\mathbf{n}\mathbf{n}^\top$; two non-parallel normals give a positive-definite sum, the integrated condition. Slide 20: stripes give one orientation inside, the checkerboard two. Intersection of constraints: Adelson and Movshon.



Slide 18. Purpose / pace: Unification; 60 seconds.

Talking points: The flow data matrix is **B**, renamed to distinguish it from the temporal **A**. The energies agree with the representation-specific residual energies up to an overall positive factor and a constant independent of flow. The common space is H^1(Omega; R^2), and the Dirichlet term couples neighboring velocities.

Explanation / transition: The notation B here is A_data in the dissertation, and alpha is alpha_smooth. The change avoids using A for two different operators on consecutive chapters. The common analysis is about fixed coefficients. A good mathematical estimator can still encode a poor representation of physical motion. Scene terms: **B** records which motion directions the edges measure and how strongly; b the measured values.

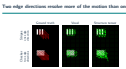


Slide 19. Purpose / pace: Central theorem; 90 seconds.

Talking points: Classical quadratic optical flow already has well-posedness theory, including Weickert and Schnörr. The contribution here is one analysis for three event-derived coefficient constructions, with explicit identifiability and perturbation hypotheses. Read the hypotheses and interpret each. Connectedness makes the kernel of the gradient consist of global constants. The positive regularization coefficient controls spatial variation. Bounded data curvature and L2 forcing make the weak forms bounded. Positive definiteness of the integrated B controls constant fields that smoothness cannot see.

Explanation / transition: B may vanish on open subsets while its integral remains positive definite. In these observation models the residual forcing b also vanishes there, and smoothness propagates information from observed regions. For time surfaces, no nonzero vector can be perpendicular to all measured gradients almost everywhere. The condition need not hold pointwise. Scene terms: edges in at least two orientations somewhere in the image. The one hypothesis the scene must supply; if it fails, a uniform slide along the null direction leaves the energy unchanged (Backup 35). It is checked on represented data, including smoothing artifacts at never-fired pixels. Accuracy rests on slide 17's shared-motion assumption (measured on slide 20).

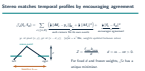
Two edge directions resolve more of the motion than one



Slide 20. Purpose / pace: Visual evidence; 60 seconds.

Talking points: The object translates diagonally. Red arrows show the ground truth and green arrows the estimates, at the common scale retained from the dissertation's panels. Vertical stripe interiors provide mainly horizontal gradients. The checkerboard provides gradient information in both spatial directions. Object boundaries can supply additional information even in the stripe scene.

Explanation / transition: These are selected illustrative windows. The full figure also includes time surface and other variants; its reported TS AEE equals the zero-field baseline at the reported precision.



Slide 21. Purpose / pace: Stereo model; about 90 seconds.

Talking points: Motivation first: a vehicle needs metric depth, how far the obstacle is, and two cameras supply the scale one camera leaves free. Frame stereo matches brightness patterns, which blur, glare, and darkness corrupt. Two event cameras watching the same edge see it change at nearly the same instant, so timing is the matching cue; WLS fits a temporal profile to each view and encourages the two profiles to agree. Rectification places corresponding rays on the same image row; disparity is the horizontal shift and similar triangles give depth.

Explanation / transition: At each candidate disparity, two softly coupled temporal profiles are solved; with frozen positive weights the Hessian is at least $\lambda_{\min}^2 \mathbf{I}$, the Chapter 2 argument with one more nonnegative term. Choosing the disparity is a separate comparison across candidates after semi-global matching aggregates the costs. Only moving edges fire, so how much of the scene receives a depth is the hard part: the next slide.

WLS scores the most points, 2.66M, at competitive depth error



Slide 22. Purpose / pace: Stereo benchmark; about 90 seconds.

Talking points: The benchmark uses the full WLS method: weight updates, cost aggregation, and candidate selection. Only moving edges fire, so how much of the scene receives a depth is the hard part, and the scored-points column is the headline.

Explanation / transition: WLS leads on Flying 2 and log RMSE; the Piatkowska adaptation leads on Flying 1, 3, and 4 by 5.9, 1.5, and 4.6 cm; WLS scores 2.66M points, the most of the five methods and 2.2 times Piatkowska's. Each method is scored on its own valid pixels, so read accuracy together with coverage. Correspondence questions go to the Chapter 4 backups.

Each next step starts from a result in the dissertation



Slide 23. Purpose / pace: Future research; about 90 seconds.

Talking points: Sensor modeling: the identity on slide 10 names the error terms; measuring them on a real sensor gives the input the stability bounds need. The GMU Neuromorphic Imaging and Digital Twins Lab (AFOSR DURIP) makes those controlled acquisitions possible. Learned regularization: $\mathbf{f}^*(\mathbf{\Lambda}) = (\mathbf{A}^\top\mathbf{A} + \mathbf{\Lambda}^2)^{-1}\mathbf{A}^\top\mathbf{y}$ is rational and hence smooth in $\mathbf{\Lambda}$ on the positive orthant, which a bilevel analysis needs from its lower level.

Explanation / transition: Shared scene: the ESAR null space contains constants and any other motion-invariant scene modes. Edge implementation: the streaming form updates each pixel in constant time; updating only pixels with events, and timing the method on a Raspberry Pi 5 with a power meter, are the next steps. Then go to the closing slide.



Slide 24. Purpose / pace: Contributions and close; about 90 seconds.

Talking points: The three panels are the estimates from slide 6. Read down each column: what the events constrain, then what the model proves. If asked about recomputed weights: Backup 65 gives a local Lipschitz bound beyond the dissertation. The temporal null mode is selected by a penalty. Spatial smoothness and integrated directional data make the flow energy coercive. Stereo has a well-posed latent solve at each candidate; selecting the candidate is a separate decision.

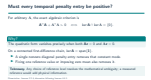
Explanation / transition: Close with the highlighted result: unique reconstruction and flow, and unique stereo inner solves under the stated hypotheses. Then stop for committee questions. Position the originality accurately: the functional-analytic tools are classical; the dissertation organizes event-derived models under explicit hypotheses and derives the coupling, conditioning, bias, and selection consequences. If asked about results beyond the table: contrast-maximization existence, graph convergence of the discrete flow, selection margins, and IRLS fixed-point and contraction results are in the backups and in Ch. 5, Table on the contribution matrix.


Discussion guide

Slide 25. Purpose / pace: Use as an appendix index.

Talking points: The following slides answer questions. PDF section bookmarks and the source-map file make it possible to find their manuscript origins.

Explanation / transition: Each discussion slide has notes that give the proof mechanism or a suggested concise answer.



Slide 26. Purpose / pace: Question: why use a full positive diagonal?

Talking points: The main theorem's positive diagonal is a strong sufficient hypothesis with an immediate quantitative lower bound. Weaker conditions also give definiteness. The kernel-intersection argument shows the minimal condition. A one-point anchoring penalty gives definiteness, but its eigenvalue bound differs from lambda_min squared when other diagonal entries are zero.

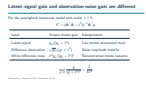
Explanation / transition: An explicit zero-mean parameterization can avoid penalizing the unobserved offset. It changes the estimator: the original magnitude penalty also attenuates nonconstant modes and controls noise. A chosen reference level fixes the estimate's offset; the absolute brightness baseline needs a measurement.



Slide 27. Purpose / pace: Question: what is the continuous-time problem?

Talking points: A finite atomic measure has a bounded-variation cumulative primitive. For a nonzero interior atom its derivative is singular, so it cannot have an L2 weak derivative and is not H1. With finite standard bounded kernels of positive width, the smoothed driver is L2 and its primitive is H1. Arbitrary L1 alone would only guarantee W1,1.

Explanation / transition: The bin convention (a,b] here matches the right-continuous process exactly. Alternative half-open bins are valid with a compatible representative or an explicit no-boundary-events assumption. The difference quotient is the integrated event increment divided by the time step; the discrete A in the presentation computes increments.



Slide 28. Purpose / pace: Question: what kind of filter is this?

Talking points: Take the singular-value decomposition A=U Sigma V^T. The reconstruction operator maps left singular modes of y to right singular modes with amplitude sqrt(g)/(g+lambda squared). Multiplying by A for noiseless y=A f_true yields the latent-signal gain g/(g+lambda squared). For white independent observation noise the output covariance diagonalizes in the same right modes and its modal variance is sigma squared times the squared observation gain.

Explanation / transition: Differentiate g/(g+a)^2 with a=lambda squared: the derivative has sign a-g, so the continuous maximum is 1/(4a) at g=a. The discrete finite-chain maximum may be smaller because its available eigenvalues need not contain a. These formulas are for a scalar penalty and equal weights.

Kernel performance depends on the scene and adaptive weights


Kernel performance depends on the scene and adaptive weights.

	Matrix	Kappa	Matrix	Kappa
Kernel	PSNR	Ch.	PSNR	Ch.
B-spline	9.66	10.15	9.72	10.20
Sinc	7.10	1470.75	7.47	52.16
Lanczos2	7.11	1470.75	7.47	52.16
Mitchell	8.69	10.53	7.66	168.34

- The temporal operator stays the same; represented data and adaptive parallax change.
- Conditioning and reconstruction live on separate properties.
- Reference image serves also measure the baseline the model adverts.

Slide 29. Purpose / pace: Question: why does a smoothing kernel change conditioning?

Talking points: The matrix K=A^T A+Lambda^T Lambda contains the bump kernel only through the observations and the weights derived from them. However Lambda is computed adaptively from represented observations. Changing aggregation can change those weights, hence the numerical matrix. The observed condition-number difference therefore comes through the data-dependent weights.

Explanation / transition: The table reports the dissertation's condition-number diagnostics. The outdoor ordering shows that kernel performance depends on the scene and adaptive weights. The positivity theorem gives a sufficient worst-case bound, independent of the empirical ranking.



Slide 30. Purpose / pace: Question: how is each metric computed, and what does it establish? About 2 minutes if all definitions are requested.

Talking points: PSNR converts the peak-squared to mean-squared pixel-error ratio to decibels. SSIM combines local means, variances and covariance to compare luminance, contrast and structure. EPI correlates mean-image gradient magnitudes with the normalized event-count map. ERC correlates absolute frame changes with absolute bump-integrated event data. TC exponentiates the negative RMS frame change divided by the reconstruction range. TG averages absolute frame changes. TV averages the sum of absolute horizontal and vertical gradients. SC averages each frame's fraction of Fourier power in the central low-frequency disk. CN divides the largest singular value of the evaluated matrix by its smallest.

Explanation / transition: Use the script for aggregation and degenerate-case details. A constant sequence has maximal TC and zero ERC under the conventions, separating smoothness from event consistency. CN describes the specified diagnostic matrix. Answer only the definitions requested; the existing slide remains an interpretation guide.

Can the reconstruction run in real time?


Can the reconstruction run in real time?

CPU measurements (Jun 2025, CPU-only build, NumPy processing time — excluding file I/O)

Mode	Resolution	Events	Batch window (s)	Batch real time	Streaming real time
1 thread/BLAS pool/numba	DAVIS346	1.0 M	0.050	2.0 s	2.1 s
	640 × 480	2.3 M	0.050	4.8 s	5.4 s
	640 × 480	2.3 M	0.100	4.8 s	5.4 s

DAVIS346: voxel grid 0.050 s, batch 2.0 s, streaming 2.1 s; cup: 2.3 M events/s, batch 4.8 s, streaming 5.4 s; 1.4 MB / 4.9 MB state; 8 threads: batch ≈ 0.8 s, streaming ≈ 1.0 s.

- DAVIS: 0.050 s voxel grid, batch 2.0 s, streaming 2.1 s.
- Streaming state: 1.4 MB (DAVIS346) / 4.9 MB (cup).
- Cup: 2.3 M events/s; batch 4.8 s, streaming 5.4 s.
- 8 threads: batch ≈ 0.8 s, streaming ≈ 1.0 s.

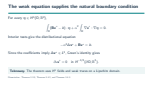
Takeaway: Only very sharp edges (DAVIS346 data, at 0.1 M ev/s) fall below 1 s with 8 threads; all other configurations remain above real time. Streaming adds ~10% overhead but enables incremental updates.

Slide 31. Purpose / pace: Question: can the reconstruction run in real time? About 60 seconds.

Talking points: These are measurements of the Chapter 2 estimator on one CPU node. DAVIS data take about four percent of real time on one thread for batch windows. The 640 by 480 cup recording at 2.3 million events per second runs at real time on one thread for batch windows and 1.3 for streaming; eight threads bring both to about a fifth of real time. Streaming state is four float32 numbers per pixel: 1.4 MB for DAVIS346 and 4.9 MB for the cup. Batch processes a complete window; streaming returns the newest endpoint of a growing prefix with finalized interval observations.

Explanation / transition: The terminal weight repeats the last interval weight, so its diagonal is one plus that weight squared. Streaming matches the full mathematical batch solve on each growing prefix; the batch code skips pixels whose net event sum is zero, so compare the two on the same pixel set. Earlier output frames can change when later data arrive. A sliding window changes the left boundary and needs additional handling. Backup 83 covers complexity, memory, and low-power choices.

The weak equation supplies the natural boundary condition



Slide 32. Purpose / pace: Question: which boundary condition is imposed?

Talking points: The energy is minimized over the full space H1, so natural Neumann conditions emerge from the first variation. Compactly supported smooth tests first give the interior equation. B v and b are in L2, hence the distributional Laplacian is L2, so the gradient has a weak normal trace.

Explanation / transition: Returning to arbitrary H1 tests, generalized Green's identity leaves a boundary pairing with their traces. Surjectivity of the trace map makes the normal trace zero. The anisotropic form replaces grad by D grad and yields a conormal condition n^T D grad v=0. No differentiability of D is needed for its weak formulation.

Slide 33. Purpose / pace: Question: how exactly does the proof control the mean? About 2 minutes.

Talking points: The gradient seminorm misses exactly the constant vector fields. Write the field as its mean plus a zero-mean part. Poincaré controls the latter. Use $\left\|\mathbf{B}^{1/2}\bar{\mathbf{v}}\right\|_{L^2} \le \left\|\mathbf{B}^{1/2}\mathbf{v}\right\|_{L^2} + \left\|\mathbf{B}^{1/2}\mathbf{w}\right\|_{L^2}$, square with the factor-two inequality, and bound the fluctuation term by $\beta_B C_P^2 \left\|\nabla\mathbf{v}\right\|_{L^2}^2$. Positive definiteness of the integrated matrix controls the mean.

Explanation / transition: An admissible lower bound is $c_0 = \min\{m/(2|\Omega|), \alpha^2/(1 + C_P^2 + 2|\Omega|\beta_B C_P^2/m)\}$. Insert it into $\|\mathbf{v}\|_{H^1}^2 = |\Omega||\bar{\mathbf{v}}|^2 + \|\mathbf{w}\|_{L^2}^2 + \|\nabla\mathbf{v}\|_{L^2}^2$. Lax–Milgram gives the unique minimizer. For fixed positive β_B, α, this lower bound deteriorates linearly as m decreases. Scene terms: mean = common drift (camera pan), fluctuation = variation across the image; smoothness charges nothing for a pan, so the edges must fix it.



Slide 34. Purpose / pace: Question: how is the stability bound derived? About 75 seconds.

Talking points: The same coercivity constant works on both coefficient pairs because the quantitative bounds are shared. Subtract their weak equations and test with the solution difference. Cauchy–Schwarz controls the right side, and $\|\mathbf{v}_2^*\|_{H^1} \le R_b/c_0$ follows from its own weak equation. Divide by $\|\mathbf{z}\|_{H^1}$ if it is nonzero.

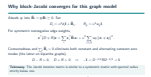
Explanation / transition: The conclusion is stability of the chosen estimator. The estimate may become poor as $m \to 0$. If a density-dependent diffusion tensor is recomputed, the regularizer also changes; its L^∞ perturbation must enter the bound. Scene terms: coefficient changes come from timestamp jitter, binning, and noise events; small m is the stripe case on slide 20.



Slide 35. Purpose / pace: Question: how is the flow computed? About 90 seconds.

Talking points: Each node stores a two-component velocity. With coefficients fixed, the normal equation at node i has the local data matrix $q_i \mathbf{B}_i$ plus alpha squared times the weighted degree. Move the neighboring velocities to the right, and use their values from the previous sweep. This gives the displayed two-by-two solve. The $q_i b_i$ term carries the local observations, and the weighted neighbor sum carries the previous surrounding field. Compute every new velocity from the same old field, then replace the whole field and repeat.

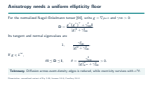
Explanation / transition: The theorem assumes a connected graph, symmetric nonnegative edge weights, positive degrees and node factors, PSD data blocks, and a positive-definite sampled Gram matrix. For this graph operator, zero graph energy forces a constant field, and the sampled data remove that constant null direction. Write $H = D - R$, where D contains the local two-by-two blocks and R contains the neighbor couplings. Both $D - R$ and $D + R$ are positive definite: possible alternating constants of the signless graph form are removed by the same sampled data condition. Hence the spectrum of the similar symmetric Jacobi matrix lies in $(-1, 1)$. This proves convergence of the simultaneous schedule from every initial field. Match a representation's stencil and update schedule to these hypotheses before applying the result. Scene terms: information spreads one neighbor per sweep.



Slide 36. Purpose / pace: Question: why does SPD suffice here for Jacobi?

Talking points: The special graph form proves more than SPD alone. For D-R, a zero edge sum forces equality of neighbors, so all node vectors equal a constant. For D+R, it forces opposite vectors on every edge; on a bipartite graph they are alternating copies of one constant vector, while an odd cycle forces zero. The sampled PSD data blocks eliminate any nonzero constant or alternating constant.

Explanation / transition: D is SPD because alpha and each degree are positive. Conjugate D plus or minus R by D^-1/2; the two positivity statements bound the symmetric spectrum between minus and plus one. D^-1R is similar to it, hence its spectral radius is below one and repeated simultaneous sweeps converge.



Slide 37. Purpose / pace: Question: can an adaptive diffusion tensor retain well-posedness?

Talking points: For a fixed, bounded, symmetric tensor with a common strictly positive lower eigenvalue bound, the Poincare/coercivity argument survives. Merely saying positive definite at each point is insufficient if eigenvalues approach zero. The normalized Nagel–Enkelmann form provides an explicit uniform floor when the density gradient is bounded.

Explanation / transition: The tensor reduces penalties on normal-direction flow variations and preserves the tangent-direction penalty. If it is recomputed after data perturbation, stability includes ||D_1-D_2||_infty. Event-density edges need not coincide with motion boundaries. The CM tensor alpha_ani squared J plus epsilon I has the opposite directional effect: it increases the normal-direction penalty.



Slide 38. Purpose / pace: Question: can you just replace the Laplacian by a tensor-weighted Hessian?

Talking points: Only for a constant tensor or when its coefficient-derivative contributions vanish. With a variable tensor the product rule supplies first-order terms. The affine test field exposes their importance particularly clearly, because its Hessian vanishes identically while its divergence-form diffusion does not.

Explanation / transition: For merely L-infinity D, those classical coefficient derivatives may not exist. Keep the weak form integral grad eta transpose D grad u; it requires no derivative of D. A flux-consistent discretization should approximate that divergence form directly.



Slide 39. Purpose / pace: Question: is your radius choice optimal?

Talking points: Differentiate the positive leading bias-squared plus variance expression, which scales as A r^4 + B r^-4. Its stationary radius obeys r^8=B/A. The resulting scaling is a guide to how timestamp noise, lattice spacing, and curvature interact.

Explanation / transition: The exact finite-design bound uses lattice sums. The asymptotic error can have either sign, so the leading surrogate approximates the bound from either side. The formula also requires positive curvature Lipschitz bound and noise, a large radius in lattice units, and inclusion in the smoothness neighborhood. Boundary truncation or sparse event support can destroy central symmetry and alter the bias.



Slide 40. Purpose / pace: Question: why distinguish the implementation from exact transport?

Talking points: Integration by parts in the OLS slope formula yields the parabolic weight w_s on the temporal derivative. Integrating the transport equation with that weight gives a correspondingly weighted spatial gradient. Using a uniform mean on one side and an OLS slope on the other generally gives a residual mismatch.

Explanation / transition: Uniform and OLS weights share zeroth and first temporal moments, so transported polynomials through degree two have zero mismatch. The cubic example supplies an actual derivative mismatch. This is a representation-coefficient error that the abstract stability theorem can address if its norm and identifiability hypotheses are controlled.



Slide 41. Purpose / pace: Question: is small-lambda ill-conditioning unavoidable?

Talking points: For this scalar unweighted system the right-hand side has zero mean in each view since A1=0, and the matrix preserves that subspace. For lambda>0 the solution lies there. Removing the constant DCT mode makes the smallest eigenvalue g_1+lambda squared instead of lambda squared, and that is positive even at lambda=0.

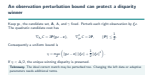
Explanation / transition: At fixed temporal dimension and coupling this gives a finite bound uniform over lambda>=0. However g_1 is asymptotic to pi squared/n squared as the unscaled temporal chain is refined, so the uniform bound grows quadratically. At a fixed positive lambda the actual restricted condition number remains finite under this unscaled refinement. The full system is singular at lambda=0 and unstructured rounding errors need not respect the mean-zero space. For positive fixed diagonal weights bounded by $W_{\max}$, $\kappa_2(\mathbf{M}_W) \le (4W_{\max} + \mu_{\max} + 2\gamma)/\mu_{\min}$, where $\mu_{\min}, \mu_{\max}$ are the extreme eigenvalues of $\mathbf{\Lambda}^2$.



Slide 42. Purpose / pace: Question: can the monocular bound be weakened?

Talking points: Complete the square in e using e_0=P^-1 S y_L. The candidate-variable center is c=y_L-e_0=P^-1 T y_L. The gap is the squared P-distance of y_d from c minus rho squared. The ideal match sits on the ellipsoid boundary. Points inside beat it, points outside lose to it, and boundary points tie.

Explanation / transition: By P-inner-product Cauchy–Schwarz the gap is at least ||e||_P(||e||_P-2rho). Therefore separation alone suffices when every mismatch exceeds twice rho. S is SPD by the Woodbury identity; T is PSD, and is SPD for gamma>0 using the full row rank of A. The characterization requires the same positive regularizer and candidate-independent coupling.



Slide 43. Purpose / pace: Question: what about noisy imperfect matches?

Talking points: The expansion of a quadratic is exact: cost change is the gradient dotted with xi plus xi transpose P xi. Bound its magnitude using ||P|| <= 1/2. Apply the uniform eta to all candidates, including the formerly ideal one, and use the elementary margin theorem.

Explanation / transition: The matrix bound follows from T <= A K^-1 A^T/4 and S=(I-A K^-1 A^T)/2, hence P <= I/2. This is an observation-level result for a fixed unweighted model. The general finite-list margin theorem applies once an appropriate score perturbation bound is independently established.



Slide 44. Purpose / pace: Question: where does the contraction constant come from?

Talking points: The difference identity isolates how changing weights changes the exact minimizer. For X=B_time^T W^1/2, M_W >= XX^T+ell I gives ||M_W^-1X|| <= ell^-1/2. The residual bound comes from comparing the frozen optimum with zero. The reciprocal identity uses ||s|-|s'|| <= |s-s'| and two bounded reciprocal weights.

Explanation / transition: This produces the epsilon^-3/2 factor after applying the weighted residual bound, then ||s-s'||_infty <= ||B_time|| ||g-g'|| and ||B_time||=||A||. The bound certifies contraction for $q < 1$. At reported $n = 20$, $\varepsilon = 0.5$, and $\lambda \in [0.01, 0.1]$, the sufficient threshold $\|\mathbf{y}\| < \lambda\varepsilon^{3/2}/[2\cos(\pi/40)]$ ranges from about 0.001773 to 0.017732. This concerns the unclipped map.



Slide 45. Purpose / pace: Question: does IRLS recover a likelihood optimum?

Talking points: At fixed variance, the log-variance term is constant and the frozen quadratic derivative is just (theta-y)/v_k. If variance is state-dependent, differentiating the likelihood also differentiates the variance and its logarithm. At theta=y the residual terms vanish but the log-variance derivative remains positive.

Explanation / transition: This simplest counterexample establishes the logical gap without depending on the actual temporal difference matrix. The dissertation also supplies an example with A=[-1,1] and positive regularization where the fixed point differs from the stationary point of the final-state-weighted score, even without the log term. Performance of the surrogate can still be evaluated empirically.



Slide 46. Purpose / pace: Question: what exactly do the experiments run?

Talking points: The implementation builds clipped solve weights, but scores at the new state with unclipped recomputed weights. A ratio cap or cost-change rule may stop the procedure before reaching a fixed point. Therefore the returned latent state is the IRLS fixed-point iterate.

Explanation / transition: Candidate guidance and uniqueness rejection insert finite penalty values before spatial averaging and SGM. Averaging mixes them with genuine costs, so a guidance window acts as a soft preference for the final winner. WTA is exact over its computed list; semi-global path aggregation is its own specified spatial algorithm.

Depth evaluation distinguishes returned coverage and scored coverage


Depth evaluation distinguishes returned coverage and scored coverage

1. Ground truth: a lane, and in the, tapering band.
2. The returned estimate, in lime.
3. The estimate also lies in the tapering band.

Depth	Returned coverage	In-band retention	Scored coverage
Control	84.47%	51.94%	1.43 M
WLS	84.48%	84.48%	2.08 M

The tapering band is [0.5, 6] m. Out-of-band: lime depths are excluded by band-based computation.

Slide 47. Purpose / pace: Question: is WLS simply much denser?

Talking points: Its returned-estimate coverage is similar to control at the reported aggregate level, and the small difference changes sign across sequences. The main scored-point difference comes from much higher estimate-side in-band retention. The benchmark excludes out-of-band estimates, so the increase in scored points is an increase in in-band coverage.

Explanation / transition: The two complete configurations differ in more than weighting, including final activity masking. The aggregate percentages use the dissertation's conventions. Raw outputs here exclude the separate posterior refinement operator but include the finest-level interpolation of the score minimum. Accuracy and retained coverage should be assessed jointly, ideally on a common population with out-of-band behavior also reported.

Exact zeros alone produce large pooled kurtosis



Slide 48. Purpose / pace: Question: why not infer a heavy-tailed likelihood from the residual plot?

Talking points: A large pooled kurtosis can be produced by many exact zeros even when the nonzero component is Gaussian. The moment calculation makes this clear. Fitted residuals also need not inherit the independence or equal variance of observation errors after regularized fitting.

Explanation / transition: This motivates an event-active analysis reporting the zero fraction, fitted-residual variance, and sequence dependence, ideally compared with synthetic residuals generated under the proposed observation model through the same fitting procedure. Robust bounded-influence penalties remain possible modeling choices.



Slide 49. Purpose / pace: Question: what do the adaptive weights do?

Talking points: Chapter 2 lowers the magnitude penalty where event activity is high. Chapter 4's empirical rule increases its weight with pre-normalization energy. The rules answer different design choices.

Explanation / transition: For differentiable parts of the Chapter 2 rules, perturbing the normal equations gives $\delta\mathbf{f}^* = \mathbf{K}^{-1}(\mathbf{A}^\top \delta\mathbf{y} - \delta(\mathbf{\Lambda}^2)\mathbf{f}^*)$. Absolute values and clipping have kinks, but every rule is still Lipschitz, which gives the local bound on page 89. The frame-assisted bilevel comparator also receives an image input, unlike event-only reconstruction.



Slide 50. Purpose / pace: Question: what is the real-data AEE?

Talking points: This real-data comparison is qualitative; the AEE values come from the known-motion examples.

Explanation / transition: The event-accumulation baseline is a separate comparison method. The theorem's coefficient representations are the time-surface, voxel, and structure-tensor constructions.


Quadratic H^1 regularization and TV answer different questions

The first energy uses a quadratic gradient penalty:

$$E(u) = \int_\Omega |\nabla u - \mathcal{M}\,\nabla u_0|^2 \, dx, \qquad u \in H^1(\Omega;\mathbb{R}^d),$$

- Lax–Milgram gives existence, uniqueness, and coefficient stability under the natural H^1 bounds.
- TV admits jump-set geometry, but needs a separate BV-existence argument and its own projection and stability assumptions.
- Normalized MI diffusion gives directional equivalent smoothing with a uniform ellipticity floor.

Takeaway. The integrated H^1 control condition supports the quadratic H^1 proof; TV needs its own existence argument.

Slide 51. Purpose / pace: Question: why use H^1 instead of TV?

Talking points: The quadratic model yields a bilinear form to which Lax–Milgram applies, then an energy identity gives coefficient stability. TV may retain discontinuities, but requires a separate existence framework in BV.

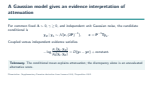
Explanation / transition: Replacing the gradient term with TV in an abstract signed-linear energy can make it unbounded below. Full-rank pointwise fidelity can retain strict convexity, so TV can keep uniqueness; with TV, H^1 coercivity needs more than the integrated condition.



Slide 52. Purpose / pace: Question: is this IRLS a majorization-minimization method?

Talking points: Only when the derivatives also match. The state-dependent variance contributes the displayed derivative term, which the frozen-weight quadratic omits; a differentiable tangent majorizer must match that derivative as well as the value.

Explanation / transition: The proved existence and conditional contraction results concern the fixed-point map. Descent is governed by the implementation's cost-rise acceptance rule, which accepts rises of up to 5%.



Slide 53. Purpose / pace: Question: can attenuation be read probabilistically?

Talking points: For fixed common parameters, interpret the magnitude and coupling penalties as a Gaussian prior and observations as independent unit-noise Gaussian measurements. The conditional distribution makes the profile prediction and its uncertainty explicit.

Explanation / transition: The prior precision is $\mathbf{M}_0 = \begin{pmatrix} \mathbf{\Lambda}^2+\gamma I & -\gamma I \\ -\gamma I & \mathbf{\Lambda}^2+\gamma I \end{pmatrix}$, and $\mathbf{M} = \mathbf{M}_0 + \operatorname{diag}(\mathbf{A}, \mathbf{A})^\top \operatorname{diag}(\mathbf{A}, \mathbf{A})$. Integrating out the profiles gives $-\log p_\gamma(\mathbf{y}) = C_\gamma(\mathbf{y}) + \frac{1}{2}\log\det\mathbf{M} - \frac{1}{2}\log\det\mathbf{M}_0 + \text{constant}$; these determinant terms are fixed across candidates. The operator $\mathbf{P}^{-1}\mathbf{T}$ has eigenvalues in $[0, 1)$ and is self-adjoint and contractive in the $\mathbf{P}$ metric, but need not contract in Euclidean norm. For scalar $\mathbf{\Lambda}$, attenuation is diagonal in the temporal observation eigenmodes. The criterion from D alone selects an ideal match for $\gamma > 0$, full-row-rank $\mathbf{A}$, and distinct candidate profiles; it is an alternative score.

The implemented estimator has several distinct layers



Slide 54. Purpose / pace: Implementation scope; 60 seconds.

Talking points: Expand IRLS as iteratively reweighted least squares and SGM as semi-global matching: SGM penalizes disparity changes along multiple paths to encourage spatial coherence. A finite search exactly returns a minimum of the final score list, subject to tie breaking. That exactness is over the computed candidate list.

Explanation / transition: Clipping affects solve weights; final scores recompute unclipped weights. Finite penalty sentinels are averaged before selection, so the guidance window is soft. The benchmark includes the finest-level interpolation of the aggregated score but excludes the separate hole-filling/refinement operator. The further analysis distinguishes the optimization and scoring steps explicitly.



Slide 55. Purpose / pace: Question: what is new if the proof tools are classical?

Talking points: Credit Tikhonov regularization, Horn–Schunck, Lax–Milgram, and Brouwer as classical tools. Explain the work as constructing and comparing precise event-based inverse models, identifying what assumptions make them well posed, and deriving consequences that are otherwise easy to overclaim. The common flow theorem gives one reusable treatment for all three representations.

Explanation / transition: The stereo cost decomposition and attenuation example explain how correct correspondence depends on more than inner solvability. The separation and margin conditions supply a conditional answer. The fixed/adaptive split prevents a quadratic guarantee from being incorrectly transferred to a changing-weight procedure. These are concrete mathematical insights that inform model design and evaluation.



Slide 56. Purpose / pace: Attribution.

Talking points: The reconstruction references distinguish learned event-only video reconstruction, flow-based image inversion, and frame-assisted bilevel deblurring. The flow references identify the classical variational structure, timestamp geometry, existing well-posedness theory, and nonconvex contrast maximization.

Explanation / transition: The dissertation's contribution is the stated event-derived models, shared coefficient analysis, and derived conditioning, bias, and selection consequences. The presentation's empirical values and theorem statements come from the dissertation. The source map ties each slide to its relevant labels, figures, and tables.



Slide 57. Purpose / pace: Question: does stability imply truth?

Talking points: No. The estimator's error splits into a shrinkage bias and propagated noise. The stability bound controls sensitivity to the observations, while the bias depends on the signal and the penalty.

Explanation / transition: A zero-mean reference fixes the unobservable constant ambiguity. On a fixed grid, noiseless data from a mean-zero reference signal are recovered as $\lambda \to 0$. Persistent noise can prevent recovery as $\lambda \to 0$; uniform refinement requires controlling noise, regularization, and grid scaling. Model error may enter the residual separately from sensor noise.

The physical event stream loses more than one baseline



Slide 58. Purpose / pace: Question: is the constant nullspace the whole sensor ambiguity?

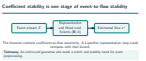
Talking points: No. The chain matrix has exactly the constant nullspace, but the sensor only fires after a contrast threshold crossing. The two displayed positive intensity signals share the same initial reference and remain inside that threshold, so both produce the same empty event stream.

Explanation / transition: This example shows that a physical stream can omit changes that never cross threshold. A small perturbation supported strictly away from existing crossings can preserve that schedule when there is a positive threshold margin.


Integrated directional information is necessary for uniqueness

Let Ω be connected, bounded, and Lipschitz, $n \in \mathbb{R}_{\geq 1}$ symmetric PSD, and $\kappa \in \mathcal{L}^1$. Minimize over symmetric PSD, $\mathbb{R}^n$-valued, $\mathbb{R}^{n \times n}$-valued, $\mathbb{R}^{n \times n}$-valued, $\mathbb{R}^{n \times n}

Coefficient stability is one stage of event-to-flow stability



Slide 60. Purpose / pace: Question: how stable is the full event-to-flow pipeline?

Talking points: The variational theorem begins after events have become coefficient fields. If preprocessing were Lipschitz in a chosen event metric, its bound could compose with the coefficient-to-flow estimate.

Explanation / transition: The metric matters: moving one positive unit atom by nonzero δt gives the total variation norm $\|\delta_{t+\delta t} - \delta_t\|_{\mathrm{TV}} = 2$, while the single-atom mass-one Wasserstein distance is $|\delta t|$. Hard bins or changing candidate neighborhoods may jump; fixed-bandwidth smoothing can behave better under a suitable metric.

Refinement requires a choice of continuum scaling



Slide 61. Purpose / pace: Question: should λ change as the grid is refined?

Talking points: The answer depends on the target continuum functional. In this smooth uniform-time illustration, difference residuals scale like h, while the Riemann sum contributes $1/h$. Matching the magnitude penalty yields $\lambda_h = \beta_{\mathrm{time}} h$.

Explanation / transition: Finite-endpoint quadrature may need endpoint weights. A physical event model would also require a scaling choice for noise and kernel bandwidth.



Slide 62. Purpose / pace: Question: what happens as α grows?

Talking points: The gradient penalty forces the minimizer toward a constant field. The data term then selects the constant that balances the integrated directional information against the integrated forcing.

Explanation / transition: For a proof, write $w_\alpha = v_\alpha - c$ and test the weak equation with it. With $h_b = b - \mathbf{B}c$ and $\int h_b = 0$, the identity is $\int w^\top \mathbf{B} w + \alpha^2 \|\nabla w\|_2^2 = \int h_b \cdot (w - \bar{w})$. Poincare gives $\|\nabla w\| = O(\alpha^{-2})$. Testing constants yields $\int \mathbf{B} w = 0$; positive definiteness of $\int \mathbf{B}$ and Poincare then give $\bar{w} \to 0$. This is a direct corollary.

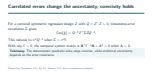


Slide 63. Purpose / pace: Question: should events be modeled as a conserved fluid?

Talking points: That is an additional physical assumption. Scalar transport follows a pattern value along a trajectory; density conservation also accounts for compression or expansion through the divergence of flow.

Explanation / transition: For a simple illustration, in one dimension let $v(x) = x$ and $V(x,t) = 1$. It solves scalar transport, but source-free conservation has residual $\partial_x(Vv) = 1$. A source term $V_t + \nabla \cdot (Vv) = s$ may be appropriate if events are created or lost.

Correlated errors change the uncertainty; coercivity holds



Slide 64. Purpose / pace: Question: what if timestamp errors are correlated?

Talking points: The covariance formula of a linear OLS estimator holds for correlated errors; the full covariance matrix replaces the white-noise simplification. If each error has the same mean, the centered symmetric design cancels that mean in the slope.

Explanation / transition: Smoothing errors with H_ϵ produces $H_\epsilon \Sigma H_\epsilon^\top$, which may be singular; check its rank before inverting. A positive definite fidelity precision Γ^{-1} keeps the regularized normal matrix positive definite, but changes coefficients and constants. A Gaussian likelihood interpretation needs a separate distributional assumption.



Slide 65. Purpose / pace: Divider for questions.

Talking points: Use when asked how time-surface gradients are estimated, or about contrast maximization, the most widely used event-based motion objective.

Explanation / transition: Contrast maximization answer in one sentence: a global minimizer exists by finite-dimensional coercivity with the coefficient ridge; the local solver carries no global guarantee.



Slide 66. Purpose / pace: Gradient-estimation result; 75 seconds; shorten to the takeaway in a compressed talk.

Talking points: The design matrix Q depends only on sample positions. Central symmetry decouples the intercept and cancels the quadratic Taylor term in the slope. A Lipschitz Hessian leaves a cubic remainder bounded by L_H |z|^3/6. Multiplication by z and Q^-1 yields the bias bound. Independent equal-variance errors give the covariance, and the MSE decomposition combines both contributions for the same discrete estimator.

Explanation / transition: A full-rank disk of timestamps around a moving straight edge can estimate its gradient well while the tangential velocity remains unidentified. The timestamp-noise model is an assumption of the theorem. The asymptotic radius heuristic is in the backup, with its finite-design qualifications.

Contrast maximization finds the motion that sharpens the events



Slide 67. Purpose / pace: What CMAX models and why it is promising; 60 seconds.

Talking points: Describe the warp in scene terms: the right motion turns a smeared trail into a sharp edge. The quadratic models first compress events into a time surface or bins and fit coefficients; CMAX keeps each event's own time. In all three known-motion scenes its AEE is the lowest, by 0.66 to 1.66 px over the structure tensor. On the stripes it recovers the diagonal, where interior pixels alone determine only the across-stripe component (normal-flow reference error 2.12 px).

Explanation / transition: Scope: one representative window (chosen by the median-deviation rule) on simulated scenes with known motion. On the idealized stripes CMAX also assigns flow to the surrounding background. The dissertation's AEE values come from the known-motion scenes. Its guarantee is existence of a global minimizer; the quasi-Newton solver is local. The next two backups give the objective and the existence proof.


How the known-motion comparison was evaluated

Label	Natural pairing
Identity	$\mathbf{W} = \mathbf{I}$, the per-event correction is the identity, so the event is left exactly where it was recorded.
Analytic	$\mathbf{W} = \mathbf{I} - \Delta t \, \mathbf{J}_v(\mathbf{x}, t)$, the first-order Taylor approximation of the backward warp.
Gradient	$\mathbf{W} = \exp(-\Delta t \, \mathbf{J}_v(\mathbf{x}, t))$, the exponential map of the same Jacobian.
Displ. / pixel	$\mathbf{W} = \mathbf{I} - \mathbf{J}_v(\mathbf{x}, t)$, the Jacobian without the Δt factor, so each vector is taken as the total displacement.

Each candidate is shown per-pixel displacement error, in pixels, on a shared color scale.

Takeaway. These numerical examples show directional differences effects on the reported values.

Dimensions, Figure 14 and Figure 15, according, see p. 1 for.

Slide 68. Purpose / pace: Question: how were the examples selected, and how fair is the comparison? About 90 seconds.

Talking points: State the motion and event generators, the shared display scale, the selection rule, and AEE units. The zero-field reference has error equal to the displacement magnitude. The settings describe the evaluated implementations; the same numerical alpha gives each implementation its own effective smoothing and computational work.

Explanation / transition: The appendix distinguishes an eight-neighbor VOX stencil from a four-neighbor structure-tensor stencil. The separate MVSEC contrast-maximization protocol has a different iteration cap. The evaluation mask is denoted by M in the cited passage; use the cited definition. Use existing real-data and CMAX backups for generalization and optimization questions.



Slide 69. Purpose / pace: Nonconvex extension; 60 seconds.

Talking points: In this model the unknown changes the represented data during optimization. q is displacement in pixels per normalized window, avoiding a confusion between velocity and normalized time. The flow is a tensor-product cubic B-spline on a uniform 8-pixel knot grid, with the two components sharing one basis. This continuous basis supplies well-defined event-location evaluations. Point evaluation of H1 functions in two dimensions needs more regularity.

Explanation / transition: The regularizer has a diffusion floor and a coefficient ridge with different roles. A positive diffusion floor penalizes all nonconstant spline variations but still leaves global translations free. The coefficient ridge controls those constant-flow directions and supplies coercivity of the full finite-dimensional objective.



Slide 70. Purpose / pace: Existence proof; 75 seconds.

Talking points: A bilinear tent contribution is a continuous function of the warped event location, including when a location crosses a pixel boundary. Each event distributes at most its absolute weight onto the canvas, so the image L1 norm and maximum absolute entry are at most W_tot. Bound variance by the second moment, then by max times L1 divided by the number of pixels.

Explanation / transition: The negative focus term is bounded below. The positive coefficient ridge drives the energy to infinity as the coefficient norm grows, so a global minimizer exists by Weierstrass. This is a finite-dimensional existence result; uniqueness and convergence of the local solver remain open. The evaluated tent objective is only piecewise differentiable, and the solver returns a local solution. The ridge eigenvalue $2\varepsilon/M$ shrinks under refinement in coefficient space; as a function-space penalty it is a uniform L^2 term (Backup 68, page 92).

Stereo depth: full Chapter 4 arc



Slide 71. Purpose / pace: Divider for questions; the full stereo arc runs about 17 minutes if presented in sequence.

Talking points: Choose the page that matches the question. For “does the correct disparity win?”, go to the counterexample and then the selection theorem. For “does IRLS converge?”, go to the fixed-point slide and its backup proof.

Explanation / transition: The next page summarizes the four results these pages prove; the core talk shows the model, the inner solve, and the MVSEC table.

Slide 72. Purpose / pace: Question: what are the Chapter 4 results? About 2 minutes; the four results and one benchmark sentence.

Talking points: This chapter compares candidate correspondences by reconstructing and softly coupling a temporal event-feature profile from each view. The analysis separates the unique inner reconstruction from the outer correspondence decision, then gives conditions for selection and stability under a positive winning margin. Item 1 is the Chapter 2 coercivity argument with one more nonnegative term. Item 2 is the conceptual point: after profiling out the latent profiles, the candidate cost contains a right-view monocular term that changes with the candidate, so in the two-sample example an attenuated wrong match beats the exact match. Item 3 gives a sufficient condition under which the correct candidate wins, and its stability under perturbations smaller than half the margin. Item 4 is the reweighted iteration.

Explanation / transition: Benchmark: each method is scored on its own valid pixels; the Piatkowska adaptation has the lowest error on Flying 1, 3, and 4. Interpret accuracy alongside scored coverage. The separation results concern unweighted candidate-local costs before aggregation; the contraction theorem applies when its displayed sufficient condition holds. Scene terms: a weaker right-view copy can outscore the true match; margin errors include noise events and sensor threshold mismatch.

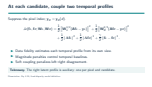


Slide 73. Purpose / pace: Question: why should corresponding observed signals agree while the rig or scene moves? About 75 seconds.

Talking points: Rectification puts simultaneous correspondences on a row. Events in each slice are then warped to a common slice end, using a separately fitted global translation in each camera. Subtract the two warped horizontal coordinates to obtain the displayed disparity equation. Closely matching per-camera warps preserve the intended row correspondence; mismatched warps alter it before optimization.

Explanation / transition: The velocities here come from the Chapter 4 preprocessing step. For originally equal vertical coordinates, their warped difference is (reference time minus event time) times the vertical-velocity difference. Global translation approximates a slice; parallax, changing disparity, or independently moving objects can introduce profile mismatch. An occluded point has no visible mate. The left-right consistency check belongs to the separate refined ablation path; the main benchmark reports raw outputs.

At each candidate, couple two temporal profiles



Slide 74. Purpose / pace: Stereo model; 75 seconds.

Talking points: The temporal operator is exactly the first-difference matrix used earlier. Lambda is a fixed positive diagonal regularizer during the candidate's iteration, and gamma is nonnegative. The right observation varies with disparity. The right latent unknown is solved anew for every left pixel and candidate. Spatial coupling enters later through the candidate-cost aggregation.

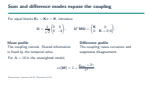
Explanation / transition: Keep the semantic distinction clear: these are uncalibrated event-feature profiles because warping, downsampling, and affine normalization transform the observations. Calling them absolute log-intensities would require a separate calibrated observation model. Unweighted theory is the case W_L=W_R=I.



Slide 75. Purpose / pace: Stereo SPD proof; 90 seconds.

Talking points: The proof reuses the temporal argument, now with an additional nonnegative coupling term. The Hessian is M_W. Every direction receives the magnitude-penalty lower bound, regardless of whether the data provide a strong temporal signal. This makes each frozen system invertible.

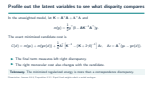
Explanation / transition: With fixed weights, subtracting normal equations gives stable dependence on the data, governed quantitatively by the inverse matrix norm and conditioning. Disparity identification is the subject of the selection theorem.



Slide 76. Purpose / pace: Spectral structure; 90 seconds.

Talking points: Adding the two normal equations eliminates coupling; subtracting them adds 2 gamma to the temporal precision. The orthonormal transform makes this exact and preserves Euclidean condition numbers. For scalar regularization, K has smallest eigenvalue lambda squared and largest g_max+lambda squared, so the displayed formula is exact.

Explanation / transition: Coupling increases the largest curvature and leaves the common constant baseline mode at lambda squared, so the full-space condition number rises with gamma at fixed lambda. The block diagonalization uses equal weights. The mean-zero restriction avoids the baseline mode for structured perturbations, but full-space conditioning remains relevant to unstructured numerical errors; see backup.



Slide 77. Purpose / pace: Profiled cost; 90 seconds.

Talking points: Complete the square in the stacked latent variables: the minimum equals ||y|| squared over two minus h^T M^-1 h over two. Use the sum–difference transform and the parallelogram identity to separate the independent monocular costs from the coupling discrepancy. The discrepancy matrix is PSD because the inverse difference is 2 gamma K^-1(K+2 gamma I)^-1.

Explanation / transition: An exact winner of these unweighted, unaggregated profiled costs also minimizes the corresponding joint latent/disparity objective. For fixed common $\mathbf{\Lambda}, \gamma$ and unit observation noise, an ideal Gaussian model with latent prior precision $\mathbf{M}_0 = \begin{pmatrix} \mathbf{\Lambda}^2+\gamma I & -\gamma I \\ -\gamma I & \mathbf{\Lambda}^2+\gamma I \end{pmatrix}$ gives profiled $C_\gamma(y)$ as $-\log p_\gamma(y)$ up to a candidate-independent constant. Its linear conditional prediction is $\mathbf{c} = \mathbf{P}^{-1}\mathbf{T}\mathbf{y}_L$; scalar-$\mathbf{\Lambda}$ modes are attenuated.

An attenuated match can cost less than the perfect one



Slide 78. Purpose / pace: Counterexample; 90 seconds.

Talking points: The example has just two temporal latent samples, so one signed difference observation per view. For each b the objective is a strictly convex quadratic and is minimized exactly. Nevertheless the perfect right observation b=1 has greater cost than b=1/2 because attenuation reduces the right monocular penalty enough to offset the discrepancy.

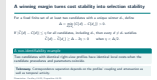
Explanation / transition: It is a rigorous counterexample to deducing correct correspondence from the unweighted inner convexity theorem. In general C(b)=(1+b^2)m(y_L)+(1-b)^2 D(y_L), so C'(1)=2m(y_L)>0 for a nonzero observation and positive regularization.



Slide 79. Purpose / pace: Selection theorem; 75 seconds.

Talking points: Under gamma>0 and full row rank of the temporal difference matrix, the discrepancy quadratic D is positive definite on the observation space. Subtract C(d*)=2m(y_L) from C(d). The monocular difference is nonnegative by assumption, and D is strictly positive for a distinct observation. Every competitor therefore loses.

Explanation / transition: This is a sufficient condition. The theorem's second separation criterion handles competitors with lower monocular costs if they are far enough away in the induced metric. Both statements assume ideal equal profiles at the physical match and concern unweighted candidate-local costs before masking or aggregation.



Slide 80. Purpose / pace: Stability and ambiguity; 75 seconds.

Talking points: A competitor's score may move down by eta while the winner's moves up by eta, so half the original margin is the sharp elementary threshold for this bound. The observation-based specialization is in the backup. This bridges continuous coefficient/data stability and the discrete argmin decision.

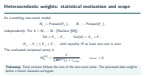
Explanation / transition: For fixed parameters, the negative log coupled-versus-independent Gaussian likelihood ratio equals the discrepancy $D = C_\gamma - C_0$ up to candidate-independent determinant terms. The discrepancy D alone favors an ideal identical profile over distinct competitors when $\gamma > 0$ and A has full row rank. Coincident right profiles still tie under identical local procedures, though spatial aggregation can create contextual differences; keep this local observation scoped to local costs.



Slide 81. Purpose / pace: Question: how does a pixel of matching error translate into centimeters? About 60 seconds.

Talking points: The conversion is reciprocal in disparity. The exact difference formula applies for positive disparities. For a small disparity perturbation, absolute depth sensitivity is Z squared divided by focal length times baseline. Farther points therefore turn the same disparity error into a larger depth error. The displayed arithmetic holds focal length and baseline fixed and is an illustration.

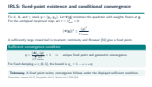
Explanation / transition: The approximation requires the disparity error to be small relative to the reference disparity; use the exact formula for the examples. Calibration uncertainty adds another source of error: to first order, delta Z/Z is delta f/f plus delta B/B minus delta d/d. A winner margin protects a label under score perturbations; subpixel interpolation and reciprocal conversion are later stages. Existing coverage backups explain method-specific evaluation populations.



Slide 82. Purpose / pace: Weight motivation; 75 seconds.

Talking points: The difference of independent Poisson counts is Skellam. In a high-count regime, a Gaussian approximation motivates weighted least squares with an inverse-variance weight. But the raw variance is the sum of polarity rates while the mean is their difference. A net-rate-dependent variance is an added approximation to the Skellam moments.

Explanation / transition: Furthermore the actual observations are real signed masses after warp, downsampling, and normalization. The weighting proxy is evaluated at the preceding state and then frozen. Its reciprocal form downweights high predicted activity relative to low activity. The reported comparison supports this rule on method-specific scored populations.



Slide 83. Purpose / pace: IRLS analysis; 90 seconds.

Talking points: Compare the frozen minimizer's objective value with the value at zero. The regularizer bounds the squared norm from below, while reciprocal weights are uniformly bounded above by 1/epsilon. Thus every closed ball with radius $R \geq \|\mathbf{y}\| / \sqrt{\ell \varepsilon_{\text{noise}}}$ is invariant. The weights are continuous and frozen matrices remain SPD, so Φ is continuous. Brouwer proves a fixed point.

Explanation / transition: A sharper difference-of-normal-equations argument gives the Lipschitz constant q. If q<1, Banach supplies uniqueness and convergence. The convergence conclusion applies under this sufficient condition; the evaluated stopping and acceptance rules are additional steps. Damping preserves fixed points. A fixed point is a stationary state of the iteration, distinct from a minimizer of the state-dependent score or a Gaussian likelihood.

Stereo benchmark: read accuracy alongside scored coverage


Stereo benchmark: read accuracy alongside scored coverage
Mean depth error (cm) ↓ log RMSE ↓
Method Flying 2 Kogler Rogister Flying 2 Kogler Rogister
Straight-line matcher 143.6 178.93 110.78 114.65 89.59 0.49
Piatkowska adaptation [1] 11.41 50.31 19.04 10.74 14.75 0.05
Xie support-block 11.41 50.31 19.04 10.74 14.75 0.05
Piatkowska adaptation [1] 37.92 74.71 33.92 12.23 16.72 0.05
Xie support-block 11.41 50.31 19.04 10.74 14.75 0.05
Rogister et al. [2] 117.61 94.61 123.47 19.51 34.62 0.19
Rogister et al. [2] 117.61 94.61 123.47 19.51 34.62 0.19
Piatkowska's local adaptation leads on most error metrics in the full dissertation table; WLS leads on Flying 2 and narrowly on log RMSE, and leads the Xie, Kogler, and Rogister rows on every mean-error column and on log RMSE.

Slide 84. Purpose / pace: Primary stereo results; 90 seconds.

Talking points: Introduce only the columns shown. Mean absolute depth error is in centimeters separately for each sequence. The displayed log RMSE is one hundred times the base metric, then averaged equally over sequence-level values. Point counts sum scored pixels across sequences. Piatkowska's local adaptation leads on most error metrics in the full dissertation table; WLS leads on Flying 2 and narrowly on log RMSE, and leads the Xie, Kogler, and Rogister rows on every mean-error column and on log RMSE.

Explanation / transition: The valid set differs by method. Both estimates and ground truth must be in the reporting depth band. A method can return many finite depths but have fewer in-band scored points. Complete configurations also differ in normalization, regularization, guidance, and final activity masking. Interpret the reported accuracy together with scored coverage and the complete method configuration.

A weighting comparison supports the selected rule


A weighting comparison supports the selected rule
Compare reciprocal sensu rata proportions' weights while holding the remaining configuration fixed.
Prop 1 Prop 2 Prop 3 Prop 4
Mean error difference (pp) 0.00 0.15 0.22 0.29
The reciprocal rule has been rejected most error on all four sequences.
Out of the sample
Reciprocal rule was chosen using these sequences, so this is an in-sample comparison.
Takeaway. The reported comparison supports the chosen rule on its scored populations; a restricted can comparison requires rescoring the outputs.
Presentation Team | Slide Supporting Document | 2026

Slide 85. Purpose / pace: Stronger empirical evidence; 60 seconds.

Talking points: The stated configurations vary the weighting rule while keeping the remaining settings fixed. Each method is scored on its own valid population. The displayed differences subtract the WLS mean errors from the WLS-Prop mean errors in the benchmark table.

Explanation / transition: The reciprocal rule was chosen using these sequences, so this is an in-sample comparison.

Hard fusion changes both the constraint and the regularization



Slide 86. Purpose / pace: Fusion result; 75 seconds.

Talking points: Set the two profiles equal in the soft energy: its two magnitude penalties add to ||Lambda v|| squared. The implemented JIF uses only half that penalty. Hence its reduced system is the infinite-gamma limit with the regularization halved. Divide its system by two to expose the effective half squared regularizer.

Explanation / transition: The mean of the soft model is already a fused profile, as the sum equation shows. The experiments compare complete configurations with different shared weights, maximum iteration counts, and scored populations.



Slide 87. Purpose / pace: Formal aperture model; about two minutes if asked.

Talking points: Here L is log-intensity. The profile ell is translated across a straight edge, with a fixed unit normal and steady illumination. Substitution into brightness constancy gives ell prime times the quantity n dot v minus s. Wherever the profile changes spatially, this determines only the normal component. Adding any constant tangential velocity produces exactly the same log-intensity history at each pixel, and therefore the same ideal threshold events from identical reset references.

Explanation / transition: For zero normal speed, translation is entirely along the ideal edge and the pixel values stay constant. This local model assumes endpoints and other distinguishing features lie outside the observation window. For nonzero normal speed, the crossing-time map has gradient n divided by s, which gives the time-surface constraint. Illumination changes are a separate source of temporal change covered by the original threshold rule.



Slide 88. Purpose / pace: Question: is there one mechanism behind the three uniqueness theorems? About 2 minutes.

Talking points: Proof: write $u = n + w$ with $n = Pu$. Since $\sqrt{a}$ is a seminorm, $a(n,n) \le 2a(u,u) + 2a(w,w) \le 2a(u,u) + 2\beta \|w\|^2$, and $\|w\|^2 \le r(u,u)/\rho$. Then $\|u\|^2 = \|n\|^2 + \|w\|^2 \le \frac{2}{\mu} a + \frac{1+2\beta/\mu}{\rho} r$. Necessity: if $N = \ker r$ and $a(n,n) = 0$ for some $n \neq 0$ in N, the form vanishes at n. Instances: temporal $\mu = \frac{1}{n}\sum_i \lambda_i^2$, $\rho = g_1 = 4\sin^2(\pi/2n)$, $\beta = \lambda_{\max}^2$; flow $\mu = m/|\Omega|$, $\rho = \alpha^2/(1 + C_P^2)$, $\beta = \beta_B$; discrete flow $\rho = \alpha^2$ times the algebraic connectivity.

Explanation / transition: Temporal: the data see every variation and miss the baseline; the penalty fixes it (Backup 02 is the exact finite-dimensional version). Flow: smoothness sees every variation and misses a uniform pan; the edges fix it (Backup 35 is the necessity statement). Stereo: with positive observation weights and $\gamma > 0$, data plus coupling leave only the common constant $(\mathbf{1}, \mathbf{1})$. Outside the lemma: correspondence selection compares finitely many candidate costs (separation margin) and contrast maximization is nonconvex (compactness gives existence only). This lemma and Lax–Milgram apply to quadratic forms; TV needs separate tools.



Slide 89. Purpose / pace: Question: the theorem fixes $\mathbf{\Lambda}$, but $\mathbf{\Lambda}$ is computed from the data. Is the actual estimator stable? About 90 seconds.

Talking points: The first term is the fixed-weight bound. The second is the weight change acting on the second solution, with $\|\mathbf{f}_2^*\| \le \|\mathbf{y}_2\| / (2\lambda_-)$. Absolute values, minima, and maxima are Lipschitz, so the kinks need no differentiability. The constant grows with $\|\mathbf{y}_2\|$, so the estimator is Lipschitz on bounded data sets; exchanging the roles of the two data sets gives the symmetric bound with the smaller norm.

Explanation / transition: The floor enters cubed, so the constants are pessimistic. The rules decrease with activity, so for data with $\|dc\|_\infty \le R$ the sigmoid floor improves to $10^{-3} + 10/(1 + e^R)$. The bound holds pixel by pixel because pixel problems are independent. Chapter 4 differs: its observation weights are recomputed from the current profiles, so the estimator is a fixed point and needs the contraction condition. Chapter 5 lists this as a future direction. The bound covers the step from the represented data to the estimate; connecting it to raw event perturbations is the representation question on slide 23.


How the parameters were chosen

Slide 90. Purpose / pace: Question: how did you choose λ, α, and γ? About 90 seconds.

Talking points: Each was chosen by grid sweep. In Chapter 2 we swept the scale of each adaptive rule, and separately the kernel type and width, then compared the three rules by image fidelity and conditioning; sigmoid has the lowest outdoor CN, 29.60. Antil and Sayre use one scalar magnitude weight; making it activity-adaptive is the change Chapter 2 evaluates. The flow smoothness weight came from grid sweeps on MVSEC, for each solver. In stereo, γ and the latent count came from a grid sweep, and inverse-variance weighting from a paired comparison, on the four evaluated sequences. The theorems hold for every positive value, so the choice moves bias and accuracy and keeps the solve unique and stable.

Explanation / transition: The discrepancy principle chooses the strongest regularization whose residual matches the noise level, so it needs that level. For raw signed counts the Skellam model gives the variance $\theta_+ + \theta_-$, estimable from ON and OFF counts; after the Chapter 4 transformation that variance must be propagated through it. The L-curve and generalized cross-validation need no noise level. The sweeps used the evaluated sequences, so a held-out protocol is the clean next experiment. The data-driven route is bilevel learning of $\mathbf{\Lambda}$ (slide 23).



Slide 91. Purpose / pace: Question: why not train a neural network? About 60 seconds.

Talking points: A trained network is also a regularized estimator. Where the events leave the answer free, it fills the gap from training images; my methods fill it with a penalty that is written down and analyzed. The dissertation identifies those free directions: the constant brightness offset at each pixel, the flow component along an edge, and disparity candidates whose temporal profiles coincide. That is also why event reconstruction has no agreed standard yet: scoring against frame images assumes a brightness level the events leave open. The approaches combine naturally: the weights $\mathbf{\Lambda}$ can be learned in a bilevel problem, and because each fixed-weight solve is unique and smooth in $\mathbf{\Lambda}$, the outer problem has a well-defined, smooth lower level. On a box of weights between a positive floor and a ceiling, a continuous outer loss attains its minimum; uniqueness and generalization need more.

Explanation / transition: If asked directly about benchmark accuracy: on the public flow and stereo benchmarks, learned methods report the lowest errors. The dissertation asks what the data determine and when each estimator is stable, and its stereo comparison is against model-based event methods. A second combination: a fixed number of block-Jacobi or IRLS iterations can be unrolled into network layers, with the convergence results describing each layer.



Slide 92. Purpose / pace: Question: the ridge $2\varepsilon/M$ vanishes as the knots are refined. Does coercivity degrade? About 90 seconds.

Talking points: The B-splines are nonnegative and sum to one, so the squared spline value is at most the Φ-weighted average of the squared coefficients. Integrating, and using that each basis function integrates to at most h^2, gives $\|u_h\|^2 \le h^2 \|\mathbf{c}\|^2$. So the ridge controls $(2\varepsilon_{\mathrm{ridge}}/(Mh^2))\,\|u_h\|^2$, and Mh^2 tends to the domain area. The eigenvalue floor controls $\beta\varepsilon_{\mathrm{ell}}\,\|\nabla u_h\|^2$. The discrete regularizer therefore controls the H^1 norm with a constant that holds for every knot spacing.

Explanation / transition: Conditioning: in two dimensions the stiffness entries are $O(\|\mathbf{D}\|_\infty)$ and each basis function overlaps at most 49 others, so Gershgorin keeps $\mu_{\max}(\mathbf{K}_0)$ bounded; with the smallest eigenvalue $\tau_{\mathrm{ridge}} \sim h^2$, $\kappa_2 = O(h^{-2})$, as for any H^1 Galerkin matrix with a mass term. Knot-spacing continuation, which the implementation supports, is the usual remedy for iteration counts. Numerical check with the clamped cubic basis: the largest one-dimensional Gram eigenvalue divided by h is 0.981, 0.994, and 1.000 for $h = 8$, 4, and 2 px. With nonuniform knots, weight each coefficient by its basis integral; the ridge is then a lumped mass term again.



Slide 93. Purpose / pace: Question: what happens as the regularization goes to zero? About 90 seconds.

Talking points: Temporal: testing the normal equations with the constant vector removes the data term at every scale, so the weighted zero-mean rule for the baseline holds for every λ. As $\lambda \to 0$ the solution converges to the minimum-weighted-norm exact fit, which Chapter 2 states. With $\mathbf{D} = \mathbf{I}$ that limit is the centered cumulative sum of the increments: noise in the increments passes through a running sum, with gain $1/\sqrt{g_1} \approx N_t/\pi$. The penalty caps the gain at $1/(2\lambda)$. Flow: the integrated condition fixes the mean, and only α controls variation. Any event-free open set gives a test field with Rayleigh quotient at most α^2. At $\alpha = 0$ the energy is pointwise and its minimizers are exactly the velocity-space lines of slide 17.

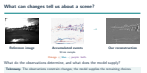
Explanation / transition: The full-space condition number $(g_{\max} + \lambda^2)/\lambda^2$ blows up only along the constant mode, which the data never excite (Backup 17 removes it). Convergence to the limit begins once λ^2 is small compared with $g_1 \approx \pi^2/N_t^2$. Positive λ and α are hypotheses of the theorems, and these limits show the role of each. For the structure tensor at $\alpha = 0$, pixels with a rank-two pooled block keep a unique velocity; the rest keep a line or the whole plane.


Where there are no events, smoothness fills in harmonically.
Let $U \subset \Omega$ be open, connected, and bipolar, with $\mathbf{B} = 0$ and $0 < a \le \nu \le M$, and $V = \{\text{all } v \in H^1 \text{ with } v = 0 \text{ on } \partial U\}$.
The weak equation forces $\nabla b = 0$ on U. The $v = b$ test component of a^* is harmonic, hence smooth, and zero on its boundary, $\int_U |\nabla b|^2 \, dx dt = 0$. Short of all $\nabla b = 0$ a.e. in U.
Theorem. We work the region contrast the lowest gradient extrapolation of the motion measured inward Ω, with no guard (top) from its boundary.

Slide 94. Purpose / pace: Question: in regions with no events, isn't the flow simply invented? About 75 seconds.

Talking points: The regularizer chooses those values, and we can say exactly how. Where the data matrix and forcing vanish, the weak equation reduces to Laplace's equation, so each velocity component is harmonic there. Equivalently, the region gets the field with least gradient energy that matches the motion on its boundary. The maximum principle bounds every directional component, and so the speed, by its values on the interface with event-active pixels. The theorem guarantees a unique regularized estimate; physical identification at each pixel needs data there.

Explanation / transition: Parts of ∂U on the image border carry the natural Neumann condition and add no constraint; the test field is admissible because test fields are free on $\partial\Omega$. For all three representations b lies in the range of $\mathbf{B}$ (residual-square data terms), so $\mathbf{B} = 0$ forces $b = 0$. The discrete analogue is the same statement for the graph Laplacian: an event-free node's velocity is the weighted average of its neighbors'. A motion boundary inside an event-free region is smoothed across; that is the H^1 modeling choice of Backup 27.



Slide 95. Purpose / pace: Motivation; about 60 seconds.

Talking points: Left is a conventional reference image of the outdoor scene. Center is a 50 ms sample of the event measurements used by the reconstruction: orange marks positive-only pixels, blue negative-only pixels, and purple pixels with both polarities. Right is our event-derived reconstruction, using a longer observation interval. The arrow represents events feeding the estimator; the reference supplies visual context. Events determine temporal changes, and the model selects a brightness baseline and how to balance observations with regularization.

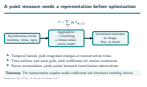
Explanation / transition: All panels use the raw left DAVIS geometry, 346×260 pixels. The 50 ms display interval ends at reconstruction time 1506118045.9162312 s and contains 23,780 events. The nearest reference is 4.672 ms later. Reconstruction uses the full 500,000-event input spanning about 0.846 s; the center panel displays a subset of that input. Event color strength is logarithmic in count and saturates at four events per pixel. The reference uses a sequence-wide min/max mapping; reconstruction uses one affine mapping across its 20-frame sequence, then clips to $[0, 1]$. This displays a latent event-derived state. The dissertation asks what observations determine, what penalties select, and what guarantees survive spatial or binocular coupling.



Slide 96. Purpose / pace: Positioning; about 90 seconds.

Talking points: For image reconstruction, Rebecq and colleagues use a recurrent neural network trained on simulated events. Zhang, Yezzi, and Gallego use optical flow to formulate a linear image inverse problem, with classical or learned image regularization. Antil and Sayre combine events with conventional frames in a bilevel problem that estimates a parameter of the event-based blur kernel. Our published temporal work reconstructs each pixel independently from event increments. I develop the temporal kernel representation and activity-based penalties, then analyze the fixed quadratic solve: what the observations determine, how regularization selects a baseline, and how perturbations affect the solution.

Explanation / transition: For flow, the starting points are Horn and Schunck's data-plus-smoothness formulation and Benosman and colleagues' event-timestamp geometry. I formulate the time-surface, voxel-projection, and structure-tensor models in a common quadratic family and identify conditions that give existence, uniqueness, and stability. Classical variational flow already has well-posedness theory, including Weickert and Schnörr's framework; this dissertation supplies the event-derived coefficient constructions and their explicit identifiability and perturbation hypotheses. Begin the models by explaining what the sensor actually reports. If asked about learned methods, use Backup 67 (page 102).



Slide 97. Purpose / pace: Representations; 60 seconds.

Talking points: A finite signed atomic measure has no ordinary density that can simply be differentiated and squared in an integral energy. Convolution, interpolation, and binning produce the quantities the models use. Raw events have distributional derivatives; the squared pointwise residuals use a smoothed representation.

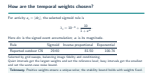
Explanation / transition: Distinguish physical intensity I, log-intensity L, event activity, and latent state f. In Chapter 2, f approximates threshold-scaled log-intensity changes; in Chapter 4, preprocessing makes f an uncalibrated event-feature profile. This distinction prevents a physical interpretation from being silently transferred between chapters.



Slide 98. Purpose / pace: Proof; 90 seconds.

Talking points: Walk through the quadratic form. Both terms are nonnegative, and the strictly positive diagonal penalty supplies a lower bound in every direction. Finite-dimensional coercivity gives existence; the positive-definite Hessian gives uniqueness. Subtracting the two normal equations gives the exact perturbation identity.

Explanation / transition: The displayed full-space bound uses $\left\|\mathbf{K}^{-1}\right\|_2 \leq \lambda_{\min}^{-2}$. A sharper operator bound is $\left\|\mathbf{K}^{-1}\mathbf{A}^\top\right\|_2 \leq 1/(2\lambda_{\min})$: from $\mathbf{A}^\top\mathbf{A} \preceq \mathbf{K} - \lambda_{\min}^2 \boldsymbol{I}$, the squared singular-value bound is at most $\max_{t \geq \lambda_{\min}^2}(1/t - \lambda_{\min}^2/t^2) = 1/(4\lambda_{\min}^2)$. The displayed estimate is a worst-case bound; the reported condition numbers describe the evaluated configurations. Scene terms: one spurious or missed event gives $\|\delta\mathbf{y}\| \leq 1$ (nonnegative unit-mass kernels), so the bound limits how far one noise event moves a history.



Slide 99. Purpose / pace: Adaptive regularization choice; 90 seconds.

Talking points: The event activity here is the absolute value of the signed accumulation in one temporal bin. Larger activity produces a smaller magnitude penalty. Sigmoid was selected after grid sweeps of each rule's scale, as a fidelity/conditioning compromise among the swept settings; Backup 66 (page 101) covers parameter selection in all three chapters. Its reported CN diagnostic is lower than the alternatives outdoors, while inverse proportional has a lower reported CN indoors.

Explanation / transition: The other two rules and their exact formulas are in the discussion page "Adaptive temporal weights are fixed in the theorem". The reported outdoor CN diagnostics are 29.60 for sigmoid, 81.50 for inverse proportional, and 100.74 for exponential decay. For any fixed positive weights, the temporal quadratic has a unique minimizer and the stability bound applies at those weights. The exact comparison identity $K_1(f_1 - f_2) = A^T(y_1 - y_2) - (\mathbf{\Lambda}_1^2 - \mathbf{\Lambda}_2^2)f_2$ shows the extra weight-change term when data-dependent weights are recomputed; with Lipschitz rules, it gives the local bound on page 89. Scene terms: quiet intervals get the largest weights and set the reference level; busy intervals get the smallest and set the worst-case noise bound.



Slide 100. Purpose / pace: Temporal results; 60 seconds.

Talking points: The pictures show a representative indoor flying 2 reference and event-only reconstruction with sigmoid weights. The table aggregates the indoor kernel comparison. PSNR measures reference-image similarity; a higher value means closer agreement. CN measures conditioning; a lower value indicates better conditioning. B-spline and Gaussian give similar PSNR and much lower CN than Delta in this indoor comparison. PSNR measures agreement with a reference frame, which includes the absolute brightness level; the events constrain changes, so reference fidelity and fidelity to the measured changes answer different questions.

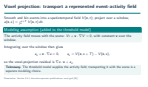
Explanation / transition: The reconstruction recovers scene structure, while the magnitude penalty selects the absolute baseline. The outdoor kernel comparison has a different ordering and appears in the backup. These results establish the observed tradeoff for the tested data and parameter choices. Display normalization is a visualization choice; the theorem's uniqueness and stability guarantees apply with the positive weights fixed.



Slide 101. Purpose / pace: Geometric derivation; 60 seconds.

Talking points: This chain-rule argument is accessible without any sensor-specific machinery. Follow a material point on the translating edge and differentiate the identity saying it stays on the zero level set.

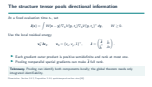
Explanation / transition: The coefficient bounds used later apply to fitted or mollified maps. The timestamp-regression result assumes a differentiable crossing-time map with Lipschitz Hessian, which are assumptions on the timestamp map. A timestamp may belong to an old edge or multiple crossings; the variational theorem bounds usable coefficient perturbations under uniform identifiability. Scene terms: $|\nabla \mathcal{T}|$ is slowness; an edge taking 10 ms per pixel moves at 100 px/s normal to itself.



Slide 102. Purpose / pace: Voxel model; 75 seconds.

Talking points: A voxel grid adds a time axis to the image plane and stores represented event contributions in time bins. The activity field measures change: a bright stationary scene has zero activity. The transport equation is a modeling assumption about this represented activity: a pixel can start or stop producing events as brightness, motion, and threshold conditions change. For an exactly transported density with constant velocity, integrate the PDE over the sliding window; this gives the displayed transport identity exactly, even for a large displacement.

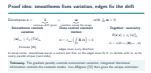
Explanation / transition: Finite temporal projection changes the profile by averaging its translates. The evaluated spatial bin mean and temporal OLS slope need not be derivative-consistent partners, and asymmetric smoothing can add mismatch. The well-posedness theorem only needs fixed bounded coefficient fields; an accurate physical-velocity interpretation also needs the transport approximation.



Slide 103. Purpose / pace: Structure tensor; 75 seconds.

Talking points: Expand the quadratic: v^T Jtilde v + 2 j_t^T v + J_tt. The last term is independent of velocity, so the spatial block is the data curvature and the temporal cross terms give the forcing. The temporal component of v_3 is fixed to one; this is an ordinary least-squares transport model.

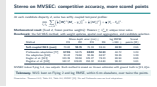
Explanation / transition: Nonnegative pooling preserves positive semidefiniteness, since z^T J z is an integral of nonnegative squared directional derivatives. A locally planar event surface is a model assumption, and smoothing scale limits temporal detail. The theorem requires only integrated positive definiteness.



Slide 104. Purpose / pace: Proof mechanism; about 90 seconds. The full proof and the stability derivation are the next two backups after the boundary-condition backup.

Talking points: The gradient seminorm misses exactly the constant vector fields. Write the field as its mean plus a zero-mean part; Poincaré controls the latter. Positive definiteness of the integrated matrix controls the mean: square $\left\|\mathbf{B}^{1/2}\bar{\mathbf{v}}\right\|_{L^2} \le \left\|\mathbf{B}^{1/2}\mathbf{v}\right\|_{L^2} + \left\|\mathbf{B}^{1/2}\mathbf{w}\right\|_{L^2}$ and bound the fluctuation term by $\beta_B C_P^2 \left\|\nabla \mathbf{v}\right\|_{L^2}^2$, $\beta_B = \left\|\mathbf{B}\right\|_{L^\infty}$. Combine the two to bound the full H^1 norm; Lax–Milgram gives existence and uniqueness.

Explanation / transition: An admissible lower bound is $c_0 = \min\{m/(2|\Omega|), \alpha^2/(1 + C_P^2 + 2|\Omega|\beta_B C_P^2/m)\}$; it deteriorates linearly as m decreases. If asked about stability (eq. abstract stability): with common bounds $\left\|\mathbf{B}_i\right\|_{L^\infty} \le \beta_B$, $\lambda_{\min}(\int \mathbf{B}_i) \ge m$, $\left\|b_i\right\|_{L^2} \le R_b$, the same c_0 gives $\left\|\mathbf{v}_1^* - \mathbf{v}_2^*\right\|_{H^1} \le c_0^{-1} \left\|b_1 - b_2\right\|_{L^2} + c_0^{-2} R_b \left\|\mathbf{B}_1 - \mathbf{B}_2\right\|_{L^\infty}$. It concerns the represented coefficients; linking it to raw sensor errors or to the true motion needs a model of the representation error. If $\int \mathbf{B}$ is singular, a constant velocity direction is uncontrolled (discussion page on necessity of integrated directional information).


Stereo on MVSEC: competitive accuracy, more scored points
All cells: mean absolute depth error, centimeters; scored coverage, percent.
$$E_{\text{data}}(w) = \sum_{(p,d)} w_p \left[\rho_{\text{data}}(I_L(p) - I_R(p-d)) + \lambda_{\text{census}} \rho_{\text{census}}(C_L(p) - C_R(p-d)) \right]$$
Weighted median (fixed w from previous weights). Reason: L_1 subproblem solved exactly.
Mean absolute depth error, centimeters; scored coverage, percent. Lower is better.
Flying 1 Flying 2 Flying 3 Flying 4
Method cm cov. cm cov. cm cov. cm cov.
Piatkowska adaptation 10.2 99.1 19.4 97.8 14.1 98.3 18.6 97.9
WLS (full method) 14.4 98.6 21.0 97.5 15.6 97.9 23.2 97.4
Piatkowska adaptation 10.2 99.1 19.4 97.8 14.1 98.3 18.6 97.9
Xie (fixed w) 24.3 99.1 40.9 97.8 29.6 98.3 42.8 97.9
Kogler (fixed w) 24.3 99.1 40.9 97.8 29.6 98.3 42.8 97.9
Rogister (fixed w) 24.3 99.1 40.9 97.8 29.6 98.3 42.8 97.9
WLS (full method) 14.4 98.6 21.0 97.5 15.6 97.9 23.2 97.4
WLS leads Flying 2 and log RMSE; Piatkowska adaptation leads Flying 1, 3, and 4 by 5.9, 1.5, and 4.6 cm.
Takeaway. WLS leads on Flying 2 and log RMSE; ranks first elsewhere; never leaves the podium.
Parameters: Census 9 × 7, $\lambda_{\text{census}} = 0.30$, ρ_{data} truncated at 0.10, ρ_{census} at 0.30, 4 iterations, 4 directions.

Slide 105. Purpose / pace: Stereo model and MVSEC results; about 2 minutes.

Talking points: One energy line, the two labels, then the table. The mathematical result is the Chapter 2 coercivity argument with one more nonnegative term, at a fixed disparity with frozen weights. The benchmark is the full method: IRLS weight updates, box-filter and SGM cost aggregation, and winner-take-all selection. Table: mean absolute depth error in centimeters per sequence; log RMSE is one hundred times the base metric, averaged over sequences; scored points are summed. WLS leads on Flying 2 and log RMSE; the Piatkowska adaptation leads on Flying 1, 3, and 4 by 5.9, 1.5, and 4.6 cm; WLS leads the Xie, Kogler, and Rogister rows on every column shown.

Explanation / transition: Each method is scored on its own valid pixels, so read accuracy together with scored coverage. In the full dissertation table the Piatkowska adaptation also leads on the median and several other error metrics. Correspondence questions (does the right candidate win? does IRLS converge?) go to the Chapter 4 backups: counterexample, separation condition, winning margin, and IRLS fixed point.



Slide 106. Purpose / pace: Close; 30 seconds.

Talking points: Return to the opening question. The dissertation gives a route from sparse asynchronous changes to deterministic estimators, and makes explicit the conditions under which each mathematical guarantee holds.

Explanation / transition: Stop here for committee questions. The appendix is organized by theorem mechanism and anticipated question.

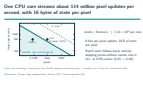


Slide 107. Purpose / pace: Question: how big is the problem, and what would run on a low-power platform? About 90 seconds.

Talking points: Independent temporal chains give O(H W N t) work for weights, right-hand sides, and tridiagonal solves. Direct indexing adds O(E k) event work. The edge path uses this structure; the main Python path uses dense solves. The example counts output storage only.

Explanation / transition: Sigmoid/delta streaming keeps three recurrence scalars plus output per pixel: O(H W) state and work per interval, matching the growing-prefix endpoint for finalized inputs. Exact savings: direct binning, buffer reuse, zero-history skipping, factor reuse for identical weights. ROI and coarser grids trade coverage/detail; shorter windows change history; reduced precision needs error checks. Earlier activity still drives quiet pixels. Symmetric kernels add look-ahead. Measure latency, peak RAM, and joules per estimate. The full script covers detection-only alternatives.

One CPU core streams about 114 million pixel updates per second, with 16 bytes of state per pixel



Slide 108. Purpose / pace: Question: how fast is it on one core, and what fits on a small board? About 2 minutes.

Talking points: Each streaming step does constant work per pixel, so the cost is one rate: about 114 million pixel updates per second on one core, 8.8 ns each. Both recordings give the same rate (114.2 and 114.5 million per second), although they differ in resolution and by nine times in event rate. Divide by the frame rate to get the pixels one core serves: about 229 thousand at 500 frames per second, 1.1 million at 100. The left point (346 by 260 pixels, 0.25 million events per second) streams at 0.39 of real time with 1.4 MB of state; the right point (640 by 480, 2.3 million events per second) at 1.3 on one core and 0.21 on eight.

Explanation / transition: All timings use one core, the closest proxy for a small board, except the eight-core line. Batch reconstruction skips pixels without events; with 6.8 percent active that is tenfold less work. If asked about the sensors: both recordings are DAVIS-class data; the rate depends on pixels and frame rate. Backup 07 adds the multithread timings; Backup 83 gives problem size and low-power options.



Slide 109. Purpose / pace: Reference list; navigation only.

Talking points: Numbers match the dissertation bibliography. Each bracketed number in the deck links here.

Explanation / transition: Return to the slide under discussion with the PDF viewer's back command.



Slide 110. Purpose / pace: Reference list; navigation only.

Talking points: Numbers match the dissertation bibliography. Each bracketed number in the deck links here.

Explanation / transition: Return to the slide under discussion with the PDF viewer's back command.



Slide 111. Purpose / pace: Reference list; navigation only.

Talking points: Numbers match the dissertation bibliography. Each bracketed number in the deck links here.

Explanation / transition: Return to the slide under discussion with the PDF viewer's back command.